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‘Project Morpheus’ - Development and Implementation of an Open-Source 3D Cellular and Organelle Morphometric Analytical Tool

Samuel Clucas, 2024.

Abstract

Accessible computational analytical tools are universally crucial in biological research in translating hypotheses into findings that deepen our understanding. In the domain of cellular biology, 3D analytical tools are being increasingly developed, where in many contexts, given the three-dimensionality of the cell, they outperform 2D tools in accuracy and validity. Such tools must be subject to critical evaluation and validation with respect to their 2D counterparts. This is naturally problematic for researchers grounded in one specific biological discipline, where consideration, and hence understanding, of concepts ranging from mathematics, computer science, physics, or chemistry are necessary. This project aims to synergise and explore concepts from a broad range of disciplines in the development of an open-source 3D morphometric analytical tool named 'Morpheus', with its implementation in analysing nuclear morphology in MDA-MB-231 breast cancer cells. The program – subject to further development, refinement, and validation – holds promise in becoming a useful tool to complement others of its kind in research and offers insight into their developmental process to encourage lateral thinking in the field.

Introduction

The diversity and dynamicity of organelle and cellular morphology in different cell types and conditions has endless implications in deepening biological understanding. Advancements in microscopy now permit the observation of cells and subcellular structures at ever improving resolutions. Confocal microscopy, cell-CT systems, 3D Structured Illumination Imaging (3D SIM) are all examples of accessible research technologies that allow the reconstruction of microscopy image stacks of cells in 3D, granting massive insights into the cellular domain.

The three-dimensionality of cell biology is easily overlooked in analytical research, where 2D analytical approaches are much more accessible and easily achieved. In certain contexts, such methods are inherently less valid than 3D analytical approaches that attempt to obtain quantitative morphometric analyses that respect the three-dimensionality of cell biology, to attain more accurate results. This is demonstrated in "Automated cell analysis in 2D and 3D: A comparative study", where cell culture and in-vivo biological understanding coupled with enhanced 3D approaches aimed to enhance early detection of lung cancer through cell classification (Meyer et al., 2009). To counter this, the rise of 3D analytical tools in biology requires careful consideration before implementation into research, where there is a "lack of validated methodologies and software tools that enable robust quantitative analysis of… 3D cellular models" (Carragher et al., 2018). There are shortcomings to 3D approaches, where poor resolution in the z-plane may impair data precision. This is a context-dependent issue, where either 2D or 3D approaches may be more appropriate, as demonstrated in a "Comparative analysis of 2D and 3D distance measurements to study spatial genome organization" (Finn et al., 2017).

The literature evidences significant differences in data obtained from the same specimen, where the dimensionality of the analytical approach varies (i.e. 2D or 3D) (Chen et al., 2011). This variability is problematic where extracting objective, quantitative, and valid data through designing an appropriate analytical pipeline is fundamental to hypothesis testing and the scientific method in biological research. This stresses the need for synergy between bioinformatics and researchers to ensure appropriate analytical techniques - both in 2D and 3D - are accessible. The amenability of analytical tools to automation is also indispensable to researchers handling large data sets, with the potential to simplify and accelerate research.

This project explores the developmental process of an open-source 3D analytical tool of cell morphology named 'Morpheus', with its implementation in observing changes to nuclear morphology in MDA-MB-231 breast cancer cell line comparing control vs. trichostatin A (TSA) treated groups. TSA is a potent histone deacetylase (HDAC) inhibitor, and hence, is known to decondense chromatin by remodelling heterochromatin. TSA has been shown to alter expression of genes implicated in apoptosis, induce cytotoxicity, induce cell cycle arrest in G1/G2 phases, amongst other anti-proliferative mechanisms; and hence cancer is one of its target pathologies (Abdelhakim Bouyahya et al., 2022). Neoplastic cancer cells exhibit aberrant cell phenotypes, where cell nuclei are often larger, multinucleate, with dramatic folds/furrows in the membrane and feature nucleoli hypertrophy (Alecsandru Ioan

Baba and Cornel Câtoi, 2024). To demonstrate the program, nuclear morphology will be analysed by 'Morpheus' for each group, and the biological implications of group differences in surface area, volume, and sphericity, explored. It should be emphasised that this is one example of its implementation, and 'Morpheus' could be implemented in its current state in cellular morphometry. This is a necessarily multidisciplinary project, incorporating well-established algorithms from computer science and mathematical theories, together with cell culture to create an accessible tool to extract meaningful data. This interdisciplinary synergy is explored, and a fundamental understanding developed, highlighting the importance of lateral thinking and cooperation in scientific research. Namely, the key concepts: random forest classifier machine-based learning image segmentation; marching cubes algorithm-based mesh generation; Divergence Theorem; the Isoperimetric Inequality, and its implication in sphericity measures (Bullard, J. and Garboczi, E. (2013)).

As mentioned, numerous projects such as this have already been developed in various contexts, establishing different analytical pipelines that achieve cell surface reconstruction, cell size measurements in 3D, spatial distribution, as well as organelle isolation and analysis, cell directivity and anisotropy (Urbaniak et al., 2022) - just to name a few use-cases. This is yet another tool to complement other available methodologies.

insightful explanation of the project

Experimental Methods

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MDA-MB-231 cells were cultured in DMEM-10% FBS. Cells were then incubated on coverslips for 24 hours in two groups: ethanol control, trichostatin A treatment.

Coverslips were fixed with 0.05% glutaraldehyde, 4% formaldehyde, 0.5% Triton X-100, and 0.1M PHEM; then blocked with PBG, stained with DAPI (an A- and T-binding fluorophore) and mounted using Vectashield.

PBG?

Coverslips were then Imaged at 60x in using the OMX superresolution microscope in conventional mode to produce deconvoluted image stacks. 30 3D cell nuclei image stacks created for each group then segmented using ILASTIK, followed by morphometric analysis by 'Morpheus'.

Exploration of Concepts Underpinning 'Morpheus' Functionality:

Program design:

Object Oriented Programming (OOP) was used to create 'Morpheus', as opposed to Procedural Oriented Programming (POP). The former implements 'classes' comprising member variables and method functions as a template for creation of an instance of a class - an 'object'.

Numerous libraries were central to its functionality, including the Visualization ToolKit (VTK) (Schroeder, Will; Martin, Ken; Lorensen, Bill, 2006) which facilitated implementation of marching cubes algorithm for mesh generation and made accessible divergence theorem through its mass properties function, now to be explored).

There are several fundamental components to the OOP paradigm: abstraction, inheritance, polymorphism, and encapsulation - all of which are featured in the program. Encapsulation reduced redundancy in the code and enhanced the robustness of program functionality to ensure broken code was isolated and easily identified and corrected without the need for whole-program correction; polymorphism facilitated the passing of data seamlessly from the Tkinter Widget object (Lundh, F., 1999) to the VTK analytical and visualisation object. Given the implementation of numerous libraries within the program, POP may have been cumbersome by comparison, where the two are compared thoroughly in A Research on Object Oriented Programming and Its Concepts, 2021.

Python was naturally the language chosen, given its ease of use and implementation, diverse libraries, and amenability to OOP (Cutting and Stephen, 2021).

Deconvolution:

Simply put, deconvolution attempts to remove image-degrading phenomena inherent to microscopy - namely "noise, scatter, glare, and blur" (comprehensively explored in Wallace, Schaefer and Swedlow, 2001). This is achieved either through deblurring algorithms which process 2D slices of a given stack or by image reconstruction algorithms which are more appropriate in correcting Point Spread Function diffraction and aberration in 3D where subsequent morphometric analysis is to be carried out. The nature of the anomaly (i.e. stochastic, or mathematically defined) and removal of anomalies through signal decomposition via the ubiquitous Fourier Transform can enhance

contrast and deblur, at the cost of decreasing the signal-to-noise ratio. When processing these image stacks, it became apparent that some artefact (likely spherical aberration) paired with this decreased signal-to-noise ratio following deconvolution resulted in visual ambiguity in fluorescing regions around the bottom and top of stacks in the z-plane (Figure 1 i), also visible in Figure 5 i) and v)), presenting issues during pre-analysis segmentation.

ILASTIK Random Forest Classifier-Based Image Segmentation:

2D slices of the 3D image stack were segmented using ILASTIK pixel classification (Stuart Berg et al., 2019) by Random Forest Classifier (RFC) to segment the nucleus/nuclear volume from background noise into a cleaned up .h5 probability file which was then thresholded using FIJI (Schindelin, J., et al., 2012) to create a binarised 3D image of cell nuclei - or more accurately, the binarised volume occupied by DAPI-stained DNA, i.e. nuclear volume - used for mesh generation in 'Morpheus'.

The RFC is a rather diplomatic supervised machine based learning algorithm with randomness intentionally woven into its decision making, resulting in an effective classifier. It can be visualised as a 'forest' of 'decision trees' central to its functionality. Each tree recursively splits the given data set at 'nodes' (the 2D slice of the volume) based on its features, ultimately classifying regions in the image as being distinct from one another. As implied, there are numerous decision trees, and each of these models follow different rules and make decisions differently - this is a relatively high-level overview of the Classification and Regression Tree (CART) algorithm by which each model is trained (Ibm.com, 2023, and visualised in Figure 1. iii)). Each tree 'votes' to classify the data, in this case labelling pixels in the image as either 'nucleus' or 'background', where the average of the votes that emerge from the forest is used to create a probability file where pixel values are assigned by the likeliness that the pixel is a part of the nuclear volume, or is simply background/noise - a prediction. Performance may be enhanced by creating larger forests consisting of 'deeper' decision trees - trees with more nodes (i.e. models that make more decisions when subsetting data). This is computationally more intensive and time consuming.

ILASTIK allows the user to train the RFC by manually labelling an image - in this case, labelling cross-sections as 'nucleus' or 'background' (third image in pipeline in Figure 1. ii) in blue, and yellow, respectively).

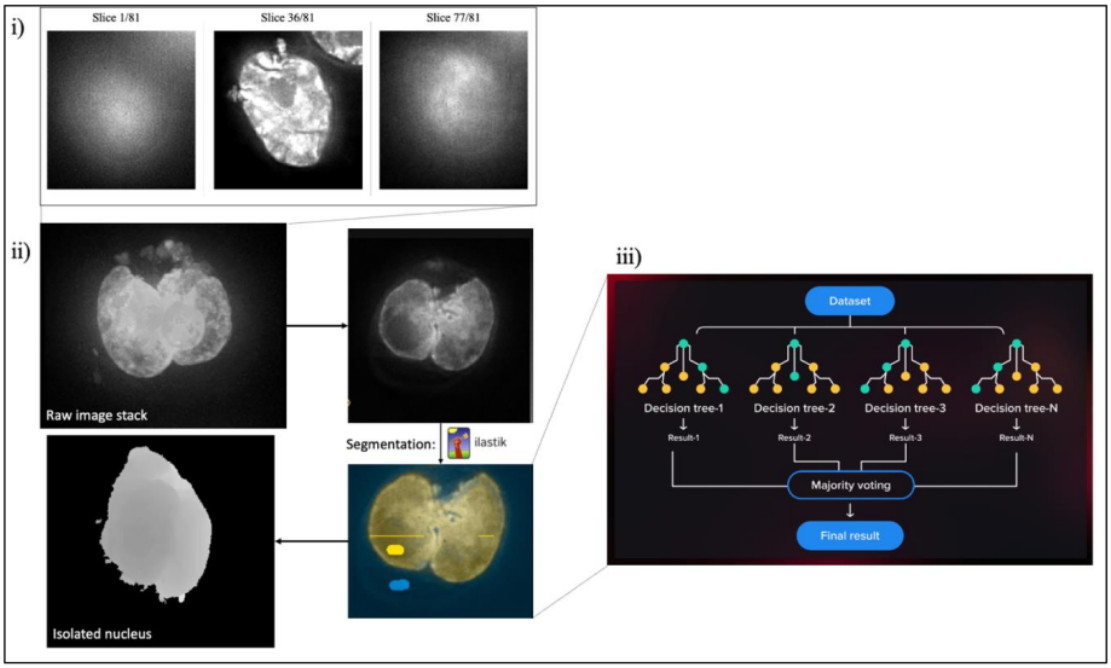


Figure 1. i) 3D image stack of one nucleus imaged, exemplifying noisy regions at the bottom and top of the stack (left and right) relative to clearer images in central regions of the stack (middle); ii) ILASTIK segmentation (Stuart Berg et al., 2019) pipeline from deconvoluted 3D image stack (top left) to segmented nucleus (bottom left) following probability file generation by supervised label-based training of Random Forest Classifier in ILASTIK and image binarization through thresholding in FIJI (Schindelin, J., et al., 2012); iii) high-level abstraction of the Random Forest Classifier machine-based learning algorithm showing input data, decision trees of the 'forest', their respective classification of the data, and majority voting, taken from Guide to Random Forest Classification and Regression Algorithms, 2022.

VTK Marching Cubes class for nuclear mesh generation:

Think of a pixel, the smallest discretised 2D unit in the xy plane, then think of a voxel as the smallest discretised 3D volumetric unit in the xyz plane like a cube, i.e. a volumetric pixel. Our 3D image is made of these voxels/cubes stored as a 3D dataset/image.

The marching cubes algorithm systematically checks each voxel, marking its vertices as either "inside" or "outside" the surface of the nuclear volume from the binarised 3D image (see Figure 2. i) for a 2D abstraction of this algorithm).

Surface reconstruction is achieved by interpolation of where the surface intersects at voxel boundaries. For each voxel (a cube with 8 vertices), the total number of possible combinations of 'inside'/'outside' vertices is 2^8, or 256. Excluding vertex configurations that are rotations or reflections of one another, there are 14 unique triangulations possible (Figure 2. ii)).

Following triangulation assignment to each voxel in the 3D image, a closed surface mesh is generated (Figure 2. iii) and iv)). Use of the class's connectivity filter ensured only the largest nuclear mesh from the 3D image input was passed to the Mass Properties class, reducing the need for more image pre-processing to isolate individual nuclei from images, improving pipeline efficiency and autonomy. This mesh is processed by VTK's mass properties class, which is used to calculate mesh surface area and volume.

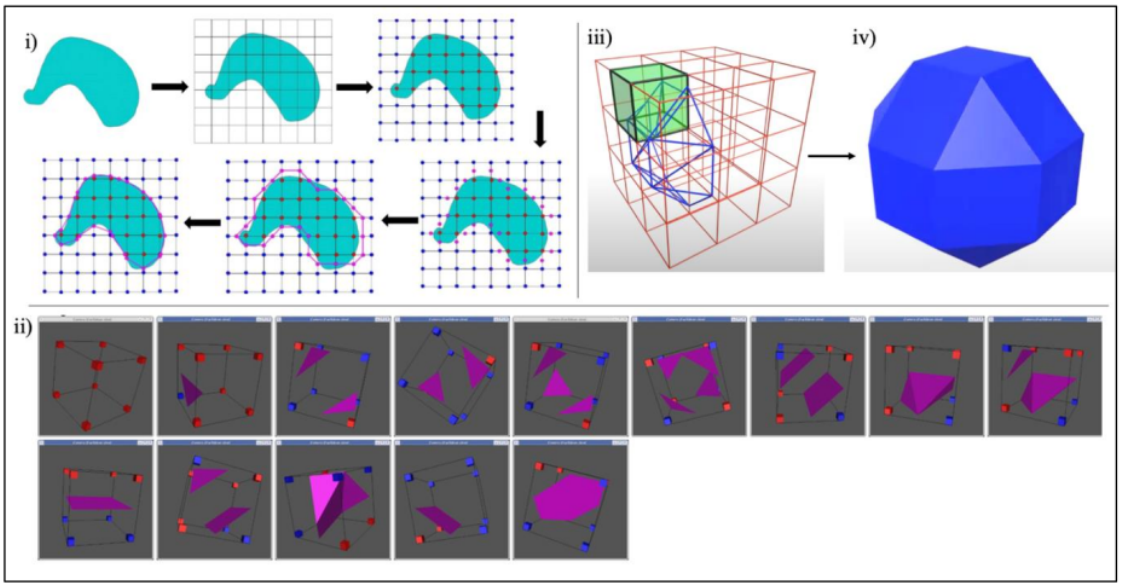


Figure 2. i) Abstraction of the Marching Cubes Algorithm in 2D: a 2d object in blue; superimposition of a square grid; labelling of corners as 'inside' (red) or 'outside' (blue) the object; interpolation of object perimeter intersections between squares in the grid by median placement of points between corners (pink points); connecting points with line segments to approximate the 2D object (pink lines); finally, movement of median points along their respective corner-connecting line segments to the actual object surface to improve the approximation (modified from Carleton.edu, 2024). ii) Renderings of the 14 unique possible surface intersections at voxel boundaries by triangulation based on 'inside' (red) or 'outside' (blue) vertices (Carleton.edu, 2024). iii) The Marching Cubes algorithm in 3D, progressively checking each voxel and its vertices' labels (the 'marching' green cube) to triangulate the surface of an object; iv) The resulting closed-surface mesh from iii) (screenshotted from Algorithms Visualized, 2017).

VTK Mass Properties class:

This class uses Gauss's Divergence Theorem from vector calculus (Equation 1), which is based on the relationship between the flux of a vector field through a closed surface (the nuclear mesh) and the divergence of the field within that surface. 'Flux' is defined as the flow and direction of a vector field through a surface. The formula is as follows:

$$\int_V (\nabla \cdot \mathbf{F})\, dV = \oint_S \mathbf{F} \cdot \hat{n}\, dS \quad (1)$$

The left side of equation 1 defines the 'divergence' of the field: the scalar product of the nabla operator (derivatives with respect to x, y, and z) and the vector field – i.e. "the sum of the derivatives of the vector field with respect to the space coordinates x, y, and z" (Physics by Alexander Fufaev, 2021). From this, the volume integral is calculated based on the field's flux into and out of each infinitesimally small volume element making up the object (Visualised on the left of Figure 3). From this, the mesh volume is calculated.

This is equal to the surface integral of the closed surface on the right of the equation, obtained through the scalar product of the vector field and the surface vector (where magnitude is surface area and direction is outwardly orthogonal to the surface - a 'normal' vector). Where the parts of the vector field for each infinitesimally small surface element is orthogonal to the surface vector, flux is 0 (given the dot product of two orthogonal vectors is 0). Hence, only parallel parts of the vector field to the surface vector will output a non-zero flux - essentially, the flow

of the vector field into and out of the closed surface. From this, mesh surface area is calculated. (Visualised on the right of Figure 3).

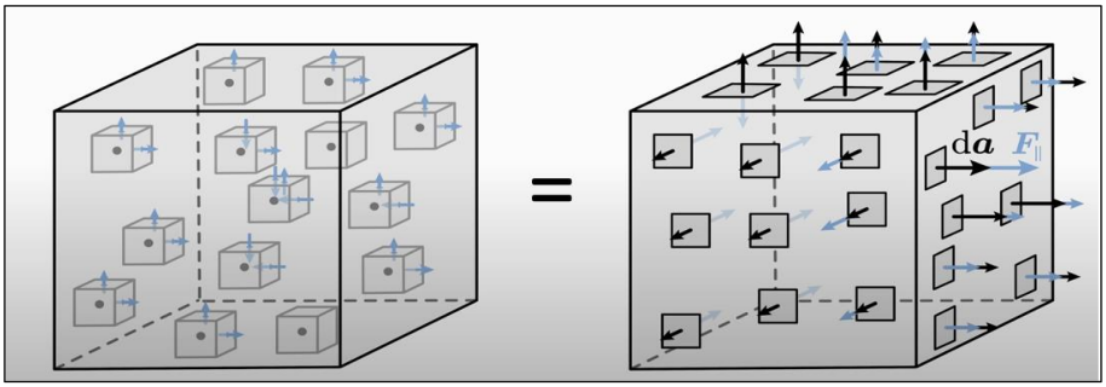


Figure 3. Visual abstractions of left and right hand side of Equation 1, where blue arrows on the left represent the flux of a vector field into and out of infinitesimally small volume elements triple integrated throughout the whole 3D cube ('sources' and 'sinks'), and on the right, blue arrows representing the flux of a vector field across the infinitesimally small surface elements of the cube parallel (blue $F_{\parallel}$) their corresponding outwardly orthogonal 'normal' vectors of said surface elements to calculate their dot product integrated over the cube's surface. (Modified from Physics by Alexander Fufaev, 2021).

Sphericity: Hakon Wadell's Definition

Sphericity, as defined in Bullard, J. and Garboczi, E. (2013), "indicates how equiaxed" an object is.

There are numerous sphericity measures based on the Isoperimetric Inequality, which states "of all bodies with the same volume, the ball has the smallest surface area" - Bullard, J. and Garboczi, E., 2013.

The program implements Hakon Wadell's definition of sphericity (Sw) - the ratio of the surface area of the object in question (Ap) to the surface area of a perfect sphere of equal volume (As). This is defined simply as:

$$Sw = \frac{A_s}{A_p} \quad (2)$$

As an object becomes increasingly spherical, the denominator approaches the numerator, where a perfect sphere has a sphericity of 1. If a perfect sphere were deformed, but its volume remained constant, its surface area to volume ratio increases, hence sphericity by this measure decreases as the denominator gets larger with respect to the numerator.

Results

Firstly, let us observe the pipeline in Figure 4 used to batch analyse the groups. Data output from ‘Morpheus’ was then subject to statistical analysis.

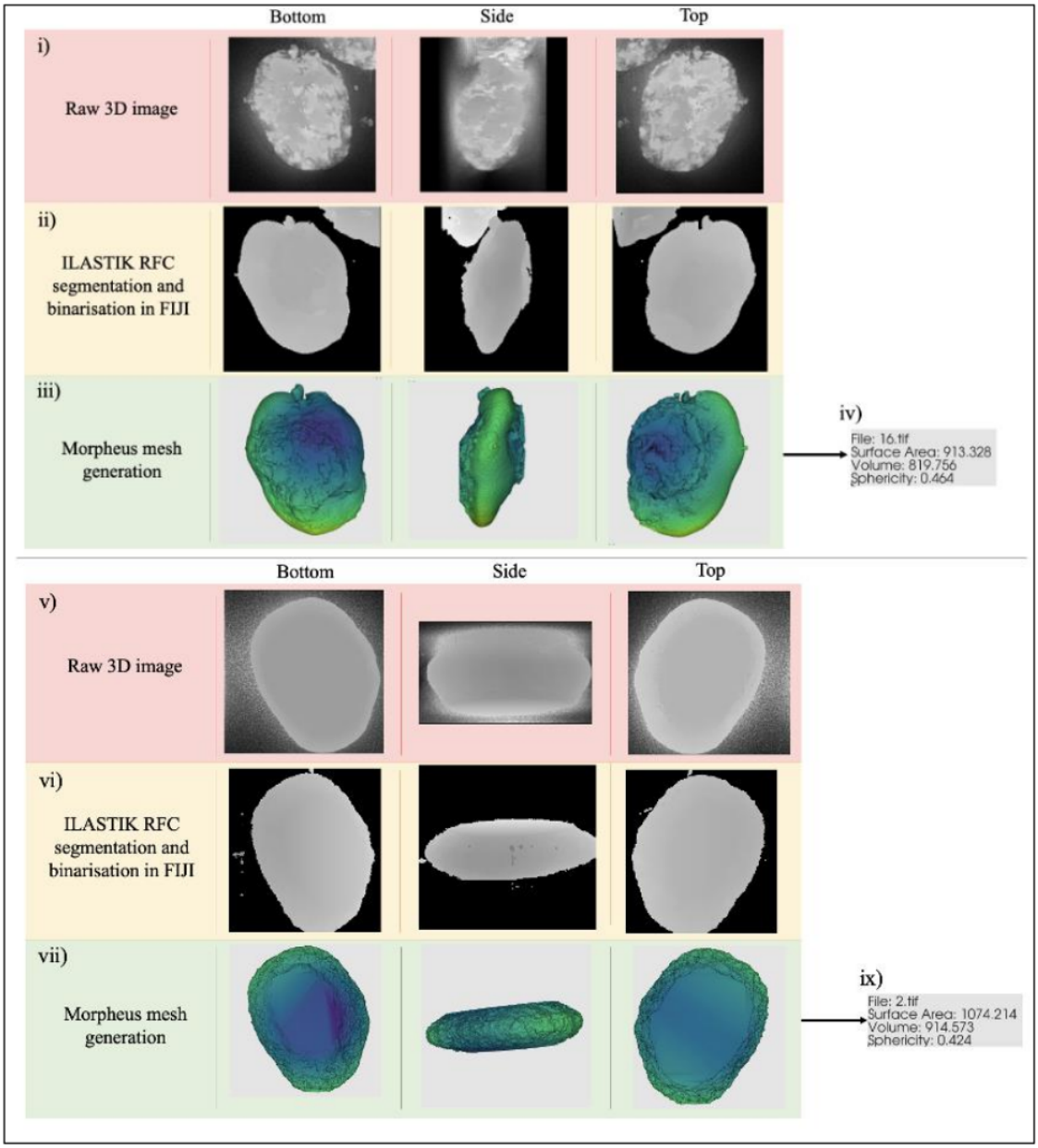


Figure 4. Pipeline overview, from deconvoluted 3D images of cell nuclei (red), to segmentation by ILASTIK and binarization by thresholding probability files in FIJI (orange), to mesh generation, visualisation (green), and parameter calculation by 'Morpheus' (iv) and ix)), where i) – iv) is one randomly selected sample 3D image from the control group, and v) – ix) is another randomly selected sample 3D image from the TSA group. 3D image projections in FIJI (pre-Morpheus processing) and meshes visualised in Morpheus are displayed from bottom, side, and top views. Morpheus mesh surfaces are coloured based on the distance from the mesh 'centroid', from close (dark blue) to far (green) where meshes exhibiting the full range of this continuous colour mapping spectrum may be considered less 'spherical'.

No difference was observed in nuclear surface area between control and TSA groups (with respective group medians of 786 voxels and 745 voxels). Cells treated with TSA had a higher average nuclear volume (with group medians of 625 voxels for the control and 805 for TSA). Sphericity was also higher in the TSA group, where the group mean for the control was 0.46, and 0.57 for TSA. (See Figure 5).

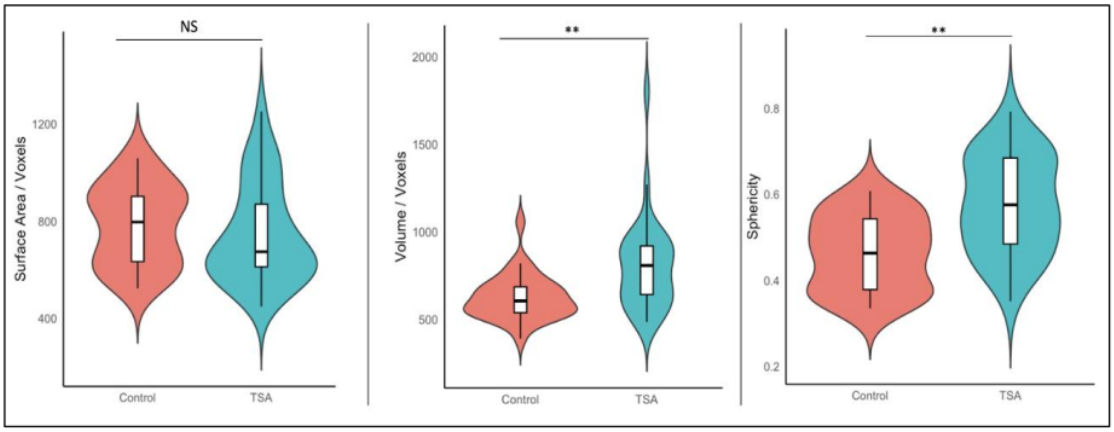


*Figure 5. Violin plots of surface area, volume, and sphericity (left to right) for control (red) and TSA (blue) groups (n=30 respectively). Surface area: no statistically significant difference (NS = P > 0.05); Wilcoxon rank-sums test, W = 515, P = 0.342. Volume: statistically significant difference (** = P < 0.01); Wilcoxon rank sums test, W = 239, P = 0.001525. Sphericity: statistically significant difference (** = P < 0.01); Welch two sample t-test, T = -4.2856, df = 54.252, P = 7.512e-05, 95% confidence interval: [-0.16443836, -0.05962831].*

Discussion

While no significant difference was found in surface area between the control and TSA groups, Lee et al., 2015 by contrast found a "significant increase in nuclear area after treatment with TSA" in MDF-7 cells". While the use of a different cell line may certainly explain a difference in response to TSA, numerous changes to the experimental approach and analytical pipeline used in this project are necessary to evaluate this.

As mentioned previously, noisy regions at the bottom and top of each 3D image stack introduced a need for individual judgement when adding labels in ILASTIK to train the RFC, diminishing pipeline objectivity. Hence, it should be emphasised that the deconvoluted and segmented 3D images have, for numerous reasons, an inherent degree of inaccuracy, and fail to delineate the precise form of the nuclei imaged. This may be diminished through the use of additional fluorophores that bind emerin-specific primary antibodies - emerin being "an integral inner

nuclear membrane protein" (Kinsella et al., 2012) - in conjunction with structured illumination microscopy (3D SIM) - which offers an improved "spatial resolution by a factor of 2 to 5" (Research Core Facilities at Drexel University, 2024). With enhanced image clarity and emerin staining, nuclear 'form' in the 3D image may be more easily discernible; variance in individual judgement during ILASTIK segmentation could be minimised, restoring pipeline objectivity. It may then be the case that data produced by 'Morpheus' would align with findings in Lee et al., 2015.

More optimistically, the result of increased nuclear volume following TSA treatment is supported in Rao et al., 2007 where HeLa cells are analysed. This is logical given the HDAC activity of TSA; chromatin remodelling and decondensing may likely be the cause for this increase in nuclear volume. This validates the pipeline to some degree, yet again, HeLa cells may respond differently, and so no conclusive statement can be made. Interestingly, however, in another study analysing the effect of TSA on MDA-MB-231 cells cultured in a 3D collagen model system, nuclei were compressed rather than expanded in contrast to other studies as well as this project (Fischer, Hayn and Claudia Tanja Mierke, 2020). This conflict in the literature highlights the need for further research into the context-dependent effects of TSA treatment on cells, not just in cancer drug research, but also in cell culture, given the use of TSA in the latter as an antifungal/antibiotic agent (Hsu et al., 2019). It also emphasises the convoluted nature of biological systems, where it's incredibly challenging to extract conclusive findings from the multifaceted and dynamic organic 'machines' that cells are.

K and Guck, 2020, found nuclear sphericity decreased post-TSA treatment, again in conflict with this project's findings, where sphericity increased following TSA treatment. Again, a different cell line was used (HeLa-FUCCI), which may respond differently to MDA-MB-231. Additionally, given the basis of Hakon Wadell's definition of sphericity on surface area, and aforementioned problems with precisely reconstructing the nuclear surface with the outlined experimental approach and pipeline, refinement in the methods used, and addition of other sphericity measures would facilitate a more comprehensive evaluation of this project's findings with respect to the literature.

Future Considerations:

With regards to future development of 'Morpheus', there is a lot to consider.

Firstly, re-writing the program in C++ would be ideal. Memory storage and access is more directly controllable, allowing for a faster-running program, capable of processing much larger data sets.

Furthermore, the marching cubes algorithm results in “blocky” meshes with steps in the surface (seen in Figure 5 iii) and vii)), reducing precision when generating a mesh to represent the nuclear volume. This is an important consideration, as the use of Divergence Theorem from VTK’s mass properties class is based on flux of a vector field across the mesh surface. Hence, if the surface reconstruction is imprecise, so will be its derived parameters.

The decreased precision in nuclear surface reconstruction warrants consideration of an alternative mathematical approach - a truncated Spherical Harmonic Series (SHS). This may achieve higher precision mesh generation and parameter calculation (nuclear surface area, volume and sphericity) (visualised in Figure 6. i)). The SHS can be used to approximate functions defined over the surface of a ‘star-shaped object’ (as defined in In Bullard and Garboczi, 2013). The star-shaped object in this case being a nucleus, with folds and furrows in their membrane. Following object surface parameterisation, series expansion, and later integration, surface area and volume may be calculated, where truncation determines precision (as in Bullard and Garboczi, 2013). The SHS is an extrapolation to 3D of the Fourier Series used to approximate periodic functions through the addition of cosine and sine functions based on their mutual orthogonality. In mathematics, this is defined as when the ‘inner product’ (the product of the definite integral of two given functions) over a given ‘interval’/period is 0. This concept may be more so intuitive when considering the basis of trigonometric functions with respect to the unit circle (figure y). One may consider sine and cosine functions as being mutually perpendicular in a mathematical sense. Theoretically speaking, the sum of an infinite series of sine and cosine functions will approach the function in question, where coefficients are determined by integrating the function of interest over a chosen interval. As the number of terms in the series increases, so does the computational cost and precision in approximating functions defined on the surface of the object - a balancing act achieved through truncation (illustrated in Figure 6. i)).

Certain phenomena such as the Gibb’s Phenomenon (exemplified in Figure 6 ii)) using a Fourier Series may also arise following the expansion of its 3D equivalent (the SHS). When the surface of the parameterised object of interest used to create the series has sharp edges or, in the case of nuclei, dramatic furrows or creases in the

membrane, such a phenomenon (described in Bullard and Garboczi, 2013 as a 'ringing artifact') may reduce precision in its approximation using the SHS. This may be resolved by increasing the order and degree of the expansion at some computational cost, yet according to the same paper 'the effect of ringing and its correction has little effect on the volume and surface area calculations' - although this may need re-validation in a biological context - hence the SHS may indeed be more efficient and precise than the algorithmic used in this project, even with minimal series expansion, warranting its implementation.

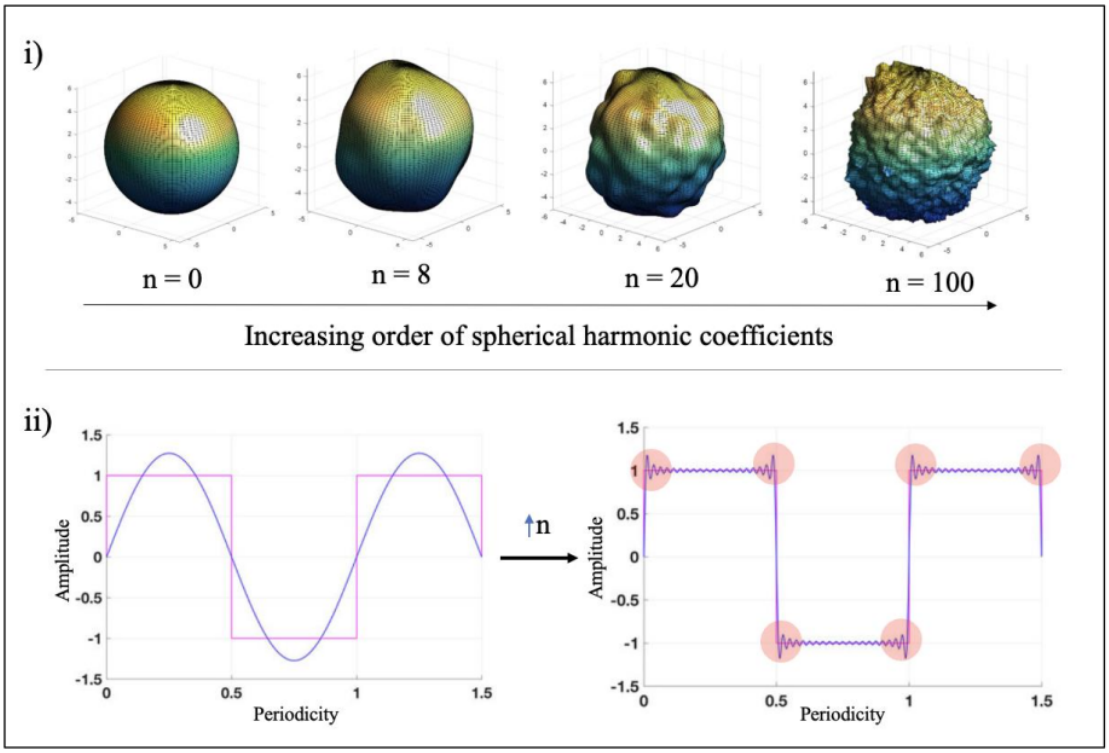


Figure 6 i) Renderings of a 3D object approximated to varying degrees of truncation (n) using the Spherical Harmonic Series (SHS) (modified from Capozza and Hanley, 2021); ii) Sinusoidal-based approximation of a square wave using a Fourier Series (lower order of coefficients on the left, higher on the right), where the Gibb's Phenomenon (observed as a 'ringing effect' in the SHS) is highlighted (red) at instantaneous transitions from the square wave function minima and maxima (modified from Research Gate, 2018).

Other sphericity measures could be added to the program to output a more comprehensive analysis. One such measure is a ratio of 'bounding spheres' (i.e. radius of maximum inscribed sphere:radius of minimum enclosing sphere), where a perfect sphere has a ratio of 1:1. Another useful measure is the 'triaxial ratio', where the primary axis is "the longest line segment that can be drawn between two surface points" (Bullard and Garboczi, 2013),

termed 'D_1'. 'D_2' is the longest line segment perpendicular to D_1, and 'D_3' is the longest line segment perpendicular to D_1 and D_2. This can be expressed as a ratio - D_1: D_2: D_3 – where a perfect sphere achieves a 1:1:1 ratio. These proposed alternative sphericity measures are useful indicators of macroscopic sphericity and complement Hakon Wadell's definition nicely given their 'surface area independence'. (Bullard and Garboczi, 2013).

There is a myriad of additional geometric measures that could be added to this 'toolkit', such as: 'Roundness' – "how blunted or rounded the corners and edges of the particle are" (Bullard and Garboczi, 2013); the opposite of roundness, 'angularity'; surface normals; 'shape entropy' – how the shape of an object affects its properties, be they physical, chemical etc., (Greg van Anders et al., 2014).

In expanding this 'toolkit', 'Morpheus' may be implemented in broader biological contexts, such as in analysing other organelles (mitochondria, vesicles, endoplasmic reticulum, etc.), as well as whole cells. Certainly, proteins and their distribution within cells could also be analysed. The applications are numerous, and so the tool itself must be versatile in its tolerance of inputs generated by different experimental protocols, as well as comprehensive in its analytical output – i.e. the program must be flexible. Programmatically, this may mean integrating a bespoke label-based RFC into the program (as in ILASTIK) to streamline research from 'wetlab' to 'drylab'. Additionally, voxel to metric conversions could be implemented to make outputted data more meaningful in a biological context, crucially enabling more direct comparison with findings in the literature by meta-analysis.

Conclusion

Early-stage development of a 3D morphometric analytical tool was achieved to some degree of success, where 60 3D images of MDA-MB-231 cell nuclei were quantitatively analysed and a pipeline established. The tool requires further development to improve its efficiency, enhance robustness and versatility, and to produce more comprehensive, meaningful data. While in its current state, it's unwise to challenge and evaluate its output with respect to the literature, certainly its further development as outlined in the future considerations above could create an accessible, useful morphometric analytical tool to ultimately help researchers go from hypothesis, to meaningful data, to advances in understanding, directing future research by the scientific method.

'Morpheus' is openly available to download with TSA and control group 3D images on Github: https://github.com/SamuelClucas/Morpheus.git

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GRADEMARK REPORT

FINAL GRADE

95 /100

GENERAL COMMENTS

First marker

This is an unusual and highly original project, the bulk of which was both self determined and self directed. It involved a huge amount of self-learning and networking with necessary experts. The first part of the project involved wet-lab cell culture, and image data generation. The rest involved development of novel software for image analysis, which was then used to analyse data from the first part. The development of the software, and the corresponding self-acquisition of skills and knowledge was a mammoth task, that we would not normally expect from a level 3 biosciences research project. The report however shows a deep understanding of the all the relevant aspects of image analysis software development, as well as the biology that was the initial aim of the project.

The Methods section includes a detailed explanation of the development of Morpheus and all the concepts that are applied to it. This section best shows the achievements of the work done, and demonstrates the ability to not only understand the background but also to creatively choose, develop and apply this understanding.

The Results section is straightforward and very clearly presented, demonstrating the outputs and showing some biologically interesting analysis, with excellent quantitative and statistical analysis.

The Discussion shows excellent derivation of information which is interpreted well and discussed in relation to published results. It is very critical and goes on to show how the work should be taken forward , also in a very critical and deeply considered way. There is an excellent balance

discussing biological results and software development that is very well integrated.

An exceptional and exemplary report, that shows a very high level of understanding, creativity, knowledge and critical thinking.

95

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2nd marker comments

This is outstanding- publishable work!

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Text Comment. insightful explanation of the project

Text Comment. PBG?

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